Structural Stagnation Probes: From Detection to Compute-Efficient Early Termination in LLM Agent Trajectories

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Abstract

LLM agents frequently enter *semantic stagnation*—executing diverse yet collectively ineffective action sequences during long-horizon tasks. We present a framework for real-time stagnation monitoring that bridges detection with compute-efficient intervention. Thirty lightweight structural features (<5 ms per window) achieve 0.967 macro-F1 in zero-shot cross-agent transfer from SWE-Agent to OpenHands, outperforming the best LLM judge by 26.6 pp in matched evaluation on identical OpenHands trajectories (0.967 vs. 0.701) and a zero-shot neural probe by 36.9 pp. Human gold validation (505 windows, 3 annotators, Fleiss κ=0.39) confirms that structural classifiers surpass majority-vote silver labels (GB 0.839 vs. MV 0.798, $p < 0.001$). On the 282 windows where labeling functions disagree (conf=1), the classifier maintains F1=0.784 versus MV agreement of 73.8%, demonstrating learned decision boundaries beyond binary heuristics. Three encoders (Qwen3-0.6B, BGE-M3, E5-large-v2) all degrade by 31–33 pp under cross-agent transfer (mean gap to structural: 35.8$\pm$1.2 pp), confirming an encoder-independent structural phenomenon. Retrospective compute-waste analysis reveals that 52.4% of agent steps occur after stagnation onset. An early termination policy at k=3 consecutive stagnation windows saves 35.3% of compute while preserving 95.5% task success, with a Pareto frontier closely tracking the oracle. Temperature-calibrated predictions achieve ECE<0.05, enabling reliable confidence-based deployment. We additionally report a well-characterized negative result: three temporal label model variants fail to improve over majority vote, constraining future research on sequential weak supervision.

1 Introduction

LLM-based autonomous agents have demonstrated impressive capabilities on software engineering tasks Yang et al. [2024], Wang et al. [2025], yet they frequently exhibit a failure mode we term *semantic stagnation*: the agent executes superficially diverse actions—different commands, different files, different strategies—that collectively make no progress toward the task goal. Unlike simple command repetition, semantic stagnation evades naïve cycle detectors because individual actions appear novel.

Recent work has quantified this degradation at scale: SlopCodeBench Anonymous [2026a] reports structural erosion in 77% of coding agent trajectories, with verbosity increasing in 75.5%. Process reward models (PRMs) Xi et al. [2026], Anonymous [2025a] address step-level evaluation but require outcome labels and RL pipelines, making them expensive training signals rather than lightweight monitors. Self-reflection approaches Shinn et al. [2023], Yao et al. [2023] rely on the stagnating model's own judgment—precisely the faculty that has failed.

We propose a framework that spans the full pipeline from detection to intervention:

1. **Structural detection dominates zero-shot transfer.** We formulate stagnation detection as window-level binary classification over 30 structural features (command repetition rates, file edit counts, output variance, etc.). A Gradient Boosting classifier trained on SWE-Agent with weak supervision achieves 0.967 cross-agent F1 on OpenHands *with zero target-domain data*, outperforming

the best LLM judge (Claude few-shot-CoT, 0.701) by 26.6 pp in matched evaluation and a zero-shot neural probe (0.598) by 36.9 pp.

2. **Gold-validated anti-circularity evidence.** Since structural features overlap with labeling function (LF) inputs, high silver-label performance risks circularity. We validate on 505 human-annotated gold windows: GB achieves 0.839 F1, exceeding MV silver labels (0.798) and the weakest human annotator (0.831). On 282 conf=1 windows where LFs disagree, GB maintains 0.784 F1 versus MV agreement of 73.8% ($p < 0.01$), proving the classifier captures beyond-LF signals.

3. **Detection-to-intervention bridge.** Retrospective analysis shows 52.4% of steps occur post-stagnation onset across 5,092 trajectories. Early termination at k=3 saves 35.3% compute while preserving 95.5% task success, with a Pareto curve closely tracking the oracle detector.

4. **Temporal label model: a well-motivated negative finding.** Three temporal label model implementations (time-weighted voting, HMM-TLM, lag-aware HMM) all fail to improve over simple majority vote, despite systematic temporal dynamics in LF behavior.

2 Related Work

Agent Failure Detection. Cycle detectors such as CDCS+CDSA and SpecRA identify structural repetition but miss semantically diverse stagnation. TrajAD detects anomalous trajectories but operates at the trajectory level rather than providing window-level monitoring. SlopCodeBench Anonymous [2026a] benchmarks coding agent degradation, reporting structural erosion in 77% of trajectories—motivating real-time detection.

Process Reward Models. AgentPRM Xi et al. [2026] trains step-level PRMs via TD/GAE to evaluate "promise and progress," but requires outcome labels and an RL pipeline. Related work includes general PRM frameworks Anonymous [2025a], ToolPRMBench Anonymous [2026b], and trajectory-level reward modeling Anonymous [2026c]. Our approach differs fundamentally: we use *window-level weak supervision* with no outcome labels, producing a monitoring tool rather than a training signal.

LLM Self-Reflection. Reflexion Shinn et al. [2023], LATS Zhou et al. [2024], and Tree of Thoughts Yao et al. [2023] leverage the agent's own reasoning for course correction. These are intervention mechanisms that depend on the stagnating model's judgment. Our structural detector is external and independent of the monitored agent.

Agent Monitoring. Recent work on monitoring LLM agents for sequentially contextual harm Anonymous [2025b] argues that sequence-level analysis is necessary—single-step evaluation is insufficient. Reliable weak-to-strong monitoring Anonymous [2025c] studies weak model supervision of strong agents for safety, complementary to our progress-focused monitoring.

Weak Supervision and Data Programming. The Snorkel paradigm Ratner et al. [2017] aggregates noisy labeling functions via generative models. We extend this to sequential agent data, discovering that temporal LF dynamics do *not* yield improved label models in the low-conflict regime (Section 6).

3 Method

3.1 Problem Formulation

Given an agent trajectory $\tau = (a_1, o_1, a_2, o_2, \ldots, a_T, o_T)$ of actions and observations, we partition τ into overlapping windows of $k = 5$ steps with stride 1. Each window $w_i = (a_i, o_i, \ldots, a_{i+4}, o_{i+4})$ is

classified as STAGNATION or PROGRESS.

3.2 Labeling Functions

We employ three labeling functions (LFs) in a weak supervision pipeline:

- **cmd_rep**: Fires when the command repetition rate within a window exceeds a threshold. High precision (97.9%) but agent-specific coverage (65.5% fire rate on SWE-Agent vs. 5.4% on OpenHands).
- **file_diff**: Fires when no file modifications are detected in the window. Cross-agent stable (fire rate 17.8% on OpenHands).
- **outcome_backpropagation (OB)**: Propagates task outcome backward as distant supervision. Privileged—requires final outcome, unavailable at inference time.

Silver labels are assigned by majority vote (MV) over LF outputs. A window receives confidence conf $\in \{1, 2, 3\}$ based on the number of agreeing LFs. For deployment, a **2-LF configuration** (cmd_rep + file_diff, no OB) achieves 0.978 F1 at conf$\geq$2, with 100% label agreement with the 3-LF configuration on the shared conf$\geq$2 subset.

3.3 Structural Features

We extract 30 structural features per window, organized into categories:

- *Command patterns*: repetition rate, unique command ratio, command diversity entropy
- *File activity*: edit binary, file access pattern, number of distinct files touched
- *Output statistics*: length variance, similarity between adjacent outputs
- *Temporal signals*: step duration trend, action type distribution
- *Error indicators*: error message count

These features are computed in $<$5 ms per window, enabling real-time monitoring.

3.4 Classifiers

We train Gradient Boosting (GB) and Random Forest (RF) classifiers on SWE-Agent silver labels (conf$\geq$2). For neural baselines, we train GRU probes on frozen embeddings from three encoders: Qwen3-Embedding-0.6B, BGE-M3, and E5-large-v2 (all $>$0.90 in-domain).

3.5 Early Termination Policy

Given the detector output $\hat{y}_i \in \{0, 1\}$ for window w_i, we define an early termination policy with threshold k: terminate execution when k consecutive windows are predicted as stagnation:

$$\text{TERMINATE} \iff \exists\, i : \hat{y}_j = 1 \;\; \forall j \in [i, i + k - 1] \tag{1}$$

4 Experiments

4.1 Anti-Circularity Validation on Gold Labels

Gold Set Construction. Three annotators labeled 505 windows sampled from SWE-Agent trajectories (stratified by confidence level), achieving Fleiss κ=0.39—consistent with definitional ambiguity in stagnation boundaries (comparable to fine-grained sentiment analysis, κ=0.40–0.55).

Category	Count	%	Pattern
GB✓ MV× (beyond-LF)	48	17.0	Subtle stag., composite signal
Both ✓	176	62.4	Clear progress
Both ×	28	9.9	Ambiguous boundary
GB× MV✓	30	10.6	FP: strategy transition

Table 1: Error decomposition on 282 conf=1 gold windows. GB rescues 48/76 (63.2%) of MV errors—the beyond-LF detection rate.

	F1	Gap to Ceiling (pp)
Adjusted Human Ceiling	0.870	—
Best Annotator (B2)	0.881	+1.1
GB (structural)	**0.839**	**−3.1**
Mean Annotator	0.853	−1.7
Weakest Annotator (B3)	0.831	−3.9
MV Silver Labels	0.798	−7.2
GRU Probe	0.790	−8.0

Table 2: Human ceiling vs. model performance on gold set (N=505). GB approaches the annotation ceiling (−3.1 pp) and exceeds the weakest individual annotator.

Conf=1 Error Decomposition. The strongest anti-circularity evidence comes from the 282 conf=1 windows where LFs disagree completely. On these windows, MV defaults to predicting progress (agreement with gold: 73.8%), while GB achieves F1=0.784 ($p < 0.01$ via paired bootstrap). Table 1 decomposes outcomes into four quadrants.

The beyond-LF detection rate of 63.2% (48 out of 76 MV errors corrected by GB) demonstrates that the classifier has learned soft decision boundaries over composite structural signals that generalize beyond binary LF heuristics. The 30 GB-wrong-MV-correct cases (10.6%) are predominantly false positives during strategy transitions, consistent with the overall FP-dominant (65%) error profile.

Human Ceiling Analysis. Table 2 contextualizes GB performance against human annotators. The Spearman-Brown adjusted ceiling is 0.870; GB at 0.839 is only 3.1 pp below, exceeding the weakest annotator (B3, 0.831).

Power Analysis. With N=505 paired bootstrap samples, we can detect $\geq$4 pp differences at α=0.05 with power=0.82. Table 3 reports power for key comparisons.

4.2 Matched LLM Comparison on OpenHands

Prior comparisons between structural classifiers and LLM judges suffered from evaluation scope confounding: structural models were evaluated on SWE-Agent while LLMs were tested on different data. We eliminate this by running LLM-as-judge (multiple prompting strategies) on the *same* OpenHands trajectories used for structural cross-agent evaluation.

Table 4 shows structural GB outperforms the best LLM (Claude few-shot-CoT) by 26.6 pp on identical data. LLMs additionally suffer cross-domain degradation: the best LLM drops 9.8 pp from SWE to OpenHands (0.799→0.701), while structural features degrade only 2.8 pp (0.995→0.967), demonstrating superior cross-domain robustness.

4.3 Encoder-Independence of the Cross-Agent Gap

A potential alternative explanation for the 36.9 pp zero-shot gap between structural features and the Qwen3-0.6B probe is encoder weakness. We test two additional encoders: BGE-M3 (BERT-based,

Comparison	Δpp	Power	Sig.?
GB vs MV	4.2	0.82	Yes (p=0.0001)
GB vs Probe	8.8	0.99	Yes (p <0.0001)
GB vs RF	2.9	0.68	Yes (p=0.0003)
TWV-Probe vs GB	−2.3	0.52	No (p=0.056)

Table 3: Power analysis for gold set comparisons (N=505, α=0.05). Effects <3 pp would require ∼1,900 samples.

Method	F1	$F1_{stag}$	$F1_{prog}$	Cost
Structural GB (OH)	**0.967**	**0.948**	**0.987**	$0
Claude few-shot-CoT	0.701	0.624	0.778	$48.3
Claude few-shot	0.662	0.581	0.743	$42.1
GPT-5.5 zero-shot	0.638	0.589	0.687	$12.8
Claude zero-shot	0.585	0.497	0.673	$13.2

Table 4: Matched comparison on OpenHands trajectories (N=2,000 stratified sample). Structural GB outperforms the best LLM variant by 26.6 pp at zero cost.

1024-d) and E5-large-v2 (contrastive-trained, 1024-d).

Despite BGE-M3 achieving +3.6 pp over Qwen3 in-domain, its zero-shot gap shrinks by only 2.3 pp (Table 6). The cross-agent gap persists across all three encoders (34.6–36.9 pp, mean 35.8±1.2 pp), confirming it reflects a fundamental signal-type difference—structural vs. semantic—rather than encoder capability.

4.4 Compute Waste Analysis

We retrospectively analyze 5,092 trajectories (1,683 SWE-Agent + 3,409 OpenHands) to quantify computation expended after the first detected stagnation onset.

Table 7 reveals that over half of all computation (52.4% of steps, 56.3% of tokens) occurs after stagnation onset. OpenHands shows higher waste (57.0% steps) due to longer trajectories and more diverse tool-calling patterns that extend ineffective exploration. The token waste exceeds step waste because stagnation-phase steps produce longer outputs (verbose error messages and repeated explanations).

4.5 Early Termination Simulation

We simulate early termination policies with consecutive-stagnation thresholds $k \in \{1, 2, 3, 5, 7\}$ using the structural GB detector on combined SWE+OH test trajectories.

The k=3 operating point (Table 8) achieves the best cost-effectiveness: 35.3% compute savings while preserving 95.5% task success. Stagnation episodes of 1–2 windows frequently self-resolve (explaining k=1's high false termination rate of 10.7%), while ≥3 consecutive stagnation windows rarely recover.

Pareto Frontier. We compare structural GB against oracle (perfect stagnation detection) and random baselines across operating points. The structural Pareto curve closely tracks the oracle (gap of only 4.0 pp at 35% savings) while dramatically outperforming random termination (+19.3 pp at the same savings level). The area under the Pareto curve is 0.891 for structural, 0.952 for oracle, and 0.623 for random—structural detection achieves 93.6% of oracle quality.

Sensitivity to Detector Quality. At k=3, a random detector saves only 22.1% compute with 81.4% success preservation, while the oracle saves 42.8% with 99.6% preservation. Structural GB (35.3% /

Variant	SWE F1	OH F1	Δpp
Claude zero-shot	0.663	0.585	−7.8
Claude few-shot-CoT	0.799	0.701	−9.8
GPT-5.5 zero-shot	0.747	0.638	−10.9

Table 5: LLM transfer degradation from SWE-Agent to OpenHands. All variants degrade 8–11 pp, versus 2.8 pp for structural features.

Encoder	In-domain	OH Zero-shot	Δpp	Gap vs Struct.
Qwen3-0.6B	0.906	0.598	−30.7	36.9 pp
BGE-M3	0.941	0.621	−32.0	34.6 pp
E5-large-v2	0.936	0.607	−32.9	36.0 pp
Structural GB	0.995†	0.967	−2.8	—

Table 6: Cross-agent zero-shot transfer by encoder. All three encoders degrade 31–33 pp; gap to structural is 35.8±1.2 pp. †Circular eval caveat: structural features overlap with LF inputs.

95.5%) sits much closer to oracle than to random, confirming that detector quality, not just the termination policy, drives the savings-success trade-off.

4.6 Deployable 2-LF Configuration

For real-time deployment, outcome_backpropagation (OB) is unavailable since task outcomes are unknown during execution. The 2-LF configuration (cmd_rep + file_diff) maintains 0.978 F1 at conf≥2 with 100% label agreement with the 3-LF configuration. However, coverage varies significantly across agents (Table 9).

The 3.6% OpenHands coverage reflects cmd_rep's low fire rate (5.4%) on OH's more diverse command patterns. This motivates a hierarchical deployment: 2-LF conf≥2 for high-confidence alerts, supplemented by the structural classifier's continuous probability output ($P(\text{stagnation}) > 0.9$) for full coverage.

4.7 Calibration Analysis

Deployment-grade early termination requires well-calibrated confidence scores. Table 10 reports expected calibration error (ECE) before and after temperature scaling.

GB predictions are highly bimodal: 64.4% of predictions exceed 0.9 and 21.7% fall below 0.1, with only 13.9% in the mid-range. Raw predictions in the 0.4–0.7 range show overconfidence (45–63% model confidence vs. 12–23% actual accuracy). Temperature scaling (T=2.06) reduces ECE from 0.061 to 0.045 on silver and from 0.083 to 0.048 on gold—both below the 0.05 deployment threshold, enabling reliable confidence-based termination decisions.

5 Analysis

5.1 Feature Importance Decomposition

SHAP analysis decomposes structural feature contributions into three categories relative to LF overlap (Table 11).

The top two features (cmd_repetition_rate: 0.142, file_edit_binary: 0.128) directly correspond to LF inputs, explaining the near-perfect silver-label performance (0.995). However, 20 residual features contribute 33.9% of total SHAP value—these features (output_length_variance, step_duration_trend, cmd_diversity_entropy, output_similarity_adjacent, action_type_distribution) have no direct LF mapping

Agent	Traj.	Steps Wasted	Tokens Wasted	Med. Tail
SWE-Agent	1,683	45.9%	49.4%	8 win.
OpenHands	3,409	57.0%	60.0%	12 win.
Combined	5,092	52.4%	56.3%	11 win.

Table 7: Compute waste by agent. 78.6% of trajectories contain at least one stagnation tail. The 90th percentile tail length is 38 windows (190 steps).

k	Prec.	Recall	Saved%	Success%	False Term.
1	0.823	0.912	46.2	89.3	10.7%
2	0.871	0.845	40.1	93.2	6.2%
3	**0.908**	**0.783**	**35.3**	**95.5**	**3.8%**
5	0.942	0.671	26.4	97.2	1.9%
7	0.961	0.582	18.8	98.1	1.1%

Table 8: Early termination simulation. k=3 is the recommended operating point: 35% savings with $<$5% task success loss and 3.8% false termination rate.

and provide the beyond-LF signal that enables GB's 0.784 F1 on conf=1 gold windows. Permutation importance ranks correlate strongly with SHAP (ρ=0.91).

5.2 Qualitative Trajectory Analysis

We present three representative trajectories illustrating key phenomena:

Example 1: Clear stagnation. An 18-window trajectory where the agent enters grep→edit→test cycling at window 7, with all three LFs firing and GB detecting immediately. This represents the "easy" case where structural repetition is directly observable.

Example 2: Semantic stagnation. A 25-window trajectory where stagnation begins at window 12, but cmd_rep does *not* fire—the agent tries different approaches (file reads, API calls, configuration changes) that are individually diverse but collectively ineffective. GB detects at window 13 via composite signal from output_length_variance and file_edit_binary. This is the core phenomenon motivating our approach: stagnation that evades repetition-based detectors.

Example 3: False positive. A 15-window trajectory where GB predicts stagnation at window 6 during a legitimate strategy transition (the agent pauses to read documentation before pivoting). Recovery occurs at window 8. This FP pattern—strategy transitions misidentified as stagnation—accounts for the majority (65%) of GB errors.

5.3 Stride Sensitivity

Window stride affects training set size, bootstrap independence, and model performance. Table 12 shows results across strides 1, 3, and 5.

Structural GB performance is virtually unchanged across strides ($<$0.4 pp), while the probe degrades by 2.4 pp at stride 5 primarily due to data reduction (7.6K vs. 38K windows). Stride 1 (default) maximizes data; stride 5 gives nearly independent windows (autocorrelation 0.11) with wider bootstrap CIs (0.042 vs. 0.018).

Agent	Windows	2-LF conf$\geq$2	Coverage
SWE-Agent	54,809	10,243	18.7%
OpenHands	150,652	5,482	3.6%

Table 9: 2-LF coverage analysis. OpenHands coverage is low due to cmd_rep's low fire rate (5.4%) on OH's diverse command patterns.

Model	Data	ECE_{raw}	ECE_{cal}	$\text{Brier}_{\text{raw}}$	$\text{Brier}_{\text{cal}}$
GB	Silver	0.061	0.045	0.063	0.059
GB	Gold	0.083	0.048	0.142	0.128
RF	Silver	0.042	0.038	0.041	0.039

Table 10: Calibration metrics. Temperature scaling (T=2.06 for GB-silver, T=1.85 for GB-gold) brings ECE below 0.05 for all configurations.

6 Temporal Label Model: A Negative Finding

LFs exhibit systematic temporal dynamics along agent trajectories: on SWE-Agent, cmd_rep leads stagnation onset by 1.87 steps, file_diff is concurrent (-1.2 steps), and OB lags by +6.7 steps. On OpenHands, cmd_rep shifts from leading to lagging (+3.32 steps). This i.i.d. violation motivated temporal label models (TLMs).

Three implementations all failed to improve over MV:

1. **Time-Weighted Voting (TWV)**: Harmful for most temporal bandwidths σ. The only marginally positive σ=2 degrades downstream classifiers: in-domain -4.9 pp, cross-agent -21.7 pp.
2. **HMM-TLM**: Overpredicts the stagnation class.
3. **Lag-aware HMM**: Merely matches MV (0.894 vs. 0.894).

The failure is attributable to the low-conflict regime: conf$\geq$2 windows (where $\geq$2 LFs agree) have 95.1% consistency with gold labels. TWV primarily changes low-confidence labels, but changes them in the wrong direction. This constitutes a well-characterized negative result: temporal LF dynamics exist but cannot be profitably exploited for label quality improvement in the current 3-LF, low-conflict setting.

7 Limitations

Agent Scope. We evaluate on two SE-domain agents (SWE-Agent, OpenHands). Generalization to non-SE agents (web browsing, scientific research) requires future validation.

Gold Set Size. N=505 is powered for $\geq$4 pp effects (power=0.82) but underpowered for smaller effects. The TWV-Probe vs. GB comparison (Δ=2.3 pp, p=0.056) is borderline due to insufficient power, not absent effect.

2-LF Coverage on OpenHands. The deployable 2-LF configuration covers only 3.6% of OpenHands windows at conf$\geq$2, due to cmd_rep's low fire rate (5.4%). Full coverage requires using the structural classifier's probability output directly, bypassing LF-based confidence.

Simulation vs. Online Deployment. Early termination results are based on offline simulation. Online deployment requires agent framework integration and may face distribution shift from the termination policy itself affecting agent behavior.

Category	#Feat.	SHAP	%	Examples
LF-overlap	2	0.270	40.1	cmd_rep_rate, file_edit
Partial-overlap	8	0.175	26.0	unique_cmd, error_count
Residual (beyond-LF)	20	0.228	33.9	output_var, duration_trend

Table 11: SHAP feature importance decomposition. Residual features (33.9%) account for GB's beyond-LF signal on conf=1 gold windows.

Stride	Train	Probe F1	GB F1	Autocorr.
1	38,161	0.906	0.995	0.72
3	12,720	0.893	0.994	0.34
5	7,632	0.882	0.992	0.11

Table 12: Stride sensitivity. Structural GB is robust ($<$0.4 pp range); probe degrades 2.4 pp due to reduced training data.

Annotator Agreement. Fleiss κ=0.39 (fair agreement) reflects genuine definitional ambiguity in stagnation boundaries, comparable to discourse relation labeling (κ=0.35–0.50). This bounds the achievable performance ceiling (estimated 0.870).

8 Conclusion

We present a framework for LLM agent stagnation monitoring spanning detection to compute-efficient intervention. Lightweight structural features achieve 0.967 cross-agent F1 with zero target-domain data, validated against human annotations (GB 0.839 vs. human ceiling 0.870, gap 3.1 pp). The beyond-LF detection rate of 63.2% on disagreement windows provides direct anti-circularity evidence. In matched evaluation, structural features outperform the best LLM judge by 26.6 pp at zero inference cost, with encoder-independence confirmed across three architectures (gap 35.8$\pm$1.2 pp). Early termination at k=3 saves 35.3% compute while preserving 95.5% task success, with a Pareto frontier at 93.6% of oracle quality. Temperature-calibrated predictions (ECE$<$0.05) and a 2-LF deployable configuration (no privileged outcome information) enable practical deployment. The negative TLM finding—temporal LF dynamics exist but cannot improve labels in low-conflict regimes—constrains future temporal weak supervision research.

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A Extended Feature Importance

The top 10 features by mean absolute SHAP value:

Feature	**SHAP**	**Category**
cmd_repetition_rate	0.142	LF-overlap
file_edit_binary	0.128	LF-overlap
unique_cmd_ratio	0.089	Partial-overlap
output_length_variance	0.067	Residual
step_duration_trend	0.054	Residual
error_message_count	0.048	Partial-overlap
cmd_diversity_entropy	0.043	Residual
file_access_pattern	0.038	Partial-overlap
output_similarity_adjacent	0.035	Residual
action_type_distribution	0.031	Residual

Table 13: Top 10 structural features by SHAP importance (N=1,824 test windows). Permutation importance ranking correlation: ρ=0.91.

B Full Pareto Frontier Data

C Stagnation Tail Length Distribution

For the 78.6% of trajectories with at least one stagnation tail:

- 25th percentile: 4 windows (20 steps)
- Median: 11 windows (55 steps)
- 75th percentile: 22 windows (110 steps)
- 90th percentile: 38 windows (190 steps)

Unresolved trajectories show substantially higher waste (58–66%) than resolved ones (10–14%), as resolved trajectories' stagnation episodes are typically brief.

Compute Saved (%)	Success Preserved (%)	Detector
18.8	98.1	Structural
26.4	97.2	Structural
35.3	95.5	Structural
40.1	93.2	Structural
46.2	89.3	Structural
25.1	99.8	Oracle
35.6	99.5	Oracle
42.8	99.1	Oracle
48.5	98.2	Oracle
52.4	96.8	Oracle
10.0	92.1	Random
20.0	84.5	Random
30.0	76.2	Random
40.0	68.1	Random
50.0	59.7	Random

Table 14: Full Pareto frontier data. Area under curve: structural 0.891, oracle 0.952, random 0.623.